

6. Applications of Integrals

6.4. Work

6.4 b Learning Objectives:

- I can understand the definitions of work and force.
- I understand how to use integration to describe work.
- I can describe the work done when pumping fluids out of containers of varying shapes.

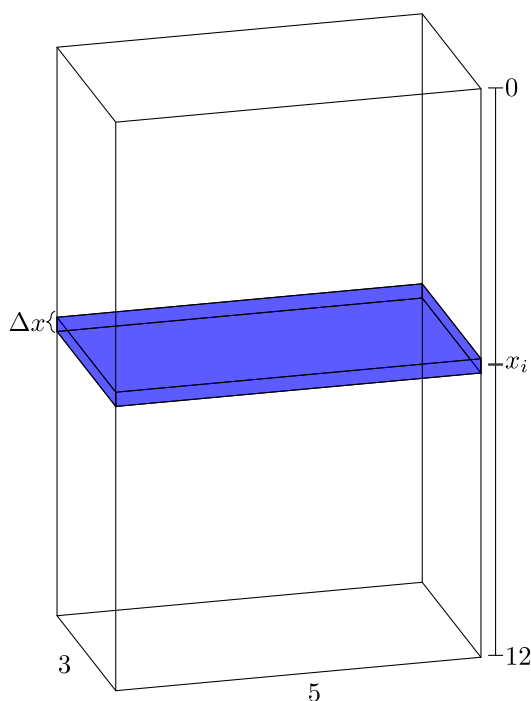
Another physical system we can calculate the work done in is when pumping liquids out of a tank.

6.4.5. Pumping Liquids

Example 6.4.1

A rectangular tank full of water has height 12 ft, width 3 ft, and length 5 ft. Find the work done in pumping all of the water to the top of the tank given that the density of water is 62.4 lbs ft^{-3} .

 **Solution**



Consider pumping a single layer of water out at a time. The work done to pump the i th layer of water out is $W_i = F_i d_i$. We will say the i th layer has to be pumped x_i ft out of the tank, and x_i goes from 0 to 12.

Then the force on each layer is the weight of the slice which can be found by multiplying density and volume: $F_i = 62.4 \text{ lbs ft}^{-3} V_i$. Now $V_i = (3)(5)(\Delta x)$, where Δx is the height of the slice. Altogether $F_i = 62.4(15)\Delta x$ lb.

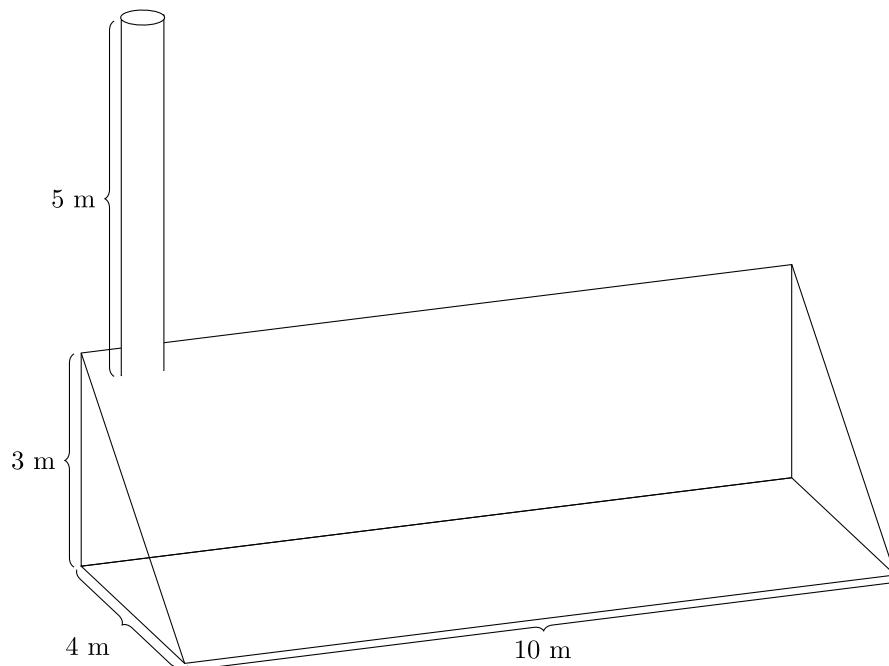
Then the work on each slice is $W_i = 62.4(15)\Delta x x_i$ lbft.

The work is then the limit of the sums of the layers

$$\begin{aligned}
 W &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (62.4)(15) \Delta x_i \\
 &= \int_0^{12} (15)(62.4)x \, dx
 \end{aligned}$$

Example 6.4.2

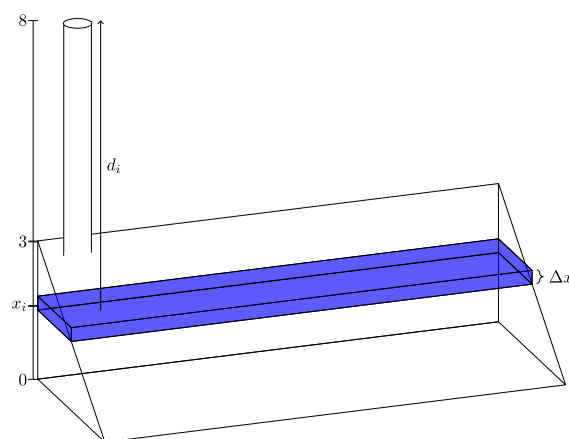
A tank pictured below is full of water. The density of water is 1000 kg m^{-3} . Set up the integral to find the work needed to pump the water to a height of 5 m above the top of the tank.



Solution

Recall that to find the work done, we consider the work to pump a generic layer of water out, then add all the layers up and take the limit as the number of layers goes to infinity to get an integral.

First, we sketch the layer



Then, the force on this layer is just gravity, so

$$\begin{aligned} F_i &= m_i a \\ &= (\text{density of water})(V_i)(g) \\ &= (1000 \text{ kg m}^{-3})(V_i)(9.8 \text{ m s}^{-2}) \end{aligned}$$

The volume of the slice is a little trickier, we need to use similar triangles to find the side length, l , of the box that touches the triangular face in terms of x_i .

We get the following relation for l .

$$\begin{aligned} \frac{4}{3} &= \frac{l}{3 - x_i} \\ l &= \frac{4}{3}(3 - x_i) \end{aligned}$$

Altogether the volume is

$$\begin{aligned} V_i &= lwh \\ &= \frac{4}{3}(3 - x_i)(10)\Delta x \end{aligned}$$

So the force to lift this layer becomes

$$F_i = (1000 \text{ kg m}^{-3})\frac{4}{3}(3 - x_i)(10)\Delta x(9.8 \text{ m s}^{-2})$$

The distance this layer has to travel to get out of the spout at the top is $d_i = 8 - x_i$, where x_i falls between $[0, 3]$ (since that is where the water is).

Thus, the work to pump a single layer is

$$W_i = (1000 \text{ kg m}^{-3})\frac{4}{3}(3 - x_i)(10)\Delta x(9.8 \text{ m s}^{-2})(8 - x_i).$$

The sum of the work for each layer gives us an approximation to the total work

$$W \approx \sum_{i=1}^n (1000 \text{ kg m}^{-3})\frac{4}{3}(3 - x_i)(10)\Delta x(9.8 \text{ m s}^{-2})(8 - x_i).$$

Then taking the limit of this sum gives us the integral for work

$$\begin{aligned} W &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (1000)\frac{4}{3}(3 - x_i)(10)\Delta x(9.8)(8 - x_i) \\ &= \int_0^3 (1000)\frac{4}{3}(3 - x)(10)(9.8)(8 - x) dx \text{ J.} \end{aligned}$$

If you want to evaluate this integral, bring constant multiples out of the integral and multiply the binomials in x then use power rule.

6.4 b Section Summary:

- We used limits of Riemann sums to describe the work done to pump liquids out of tanks.
- We had to review some formulas for some geometry to finish these problems, like similar triangles and Pythagorean Theorem.